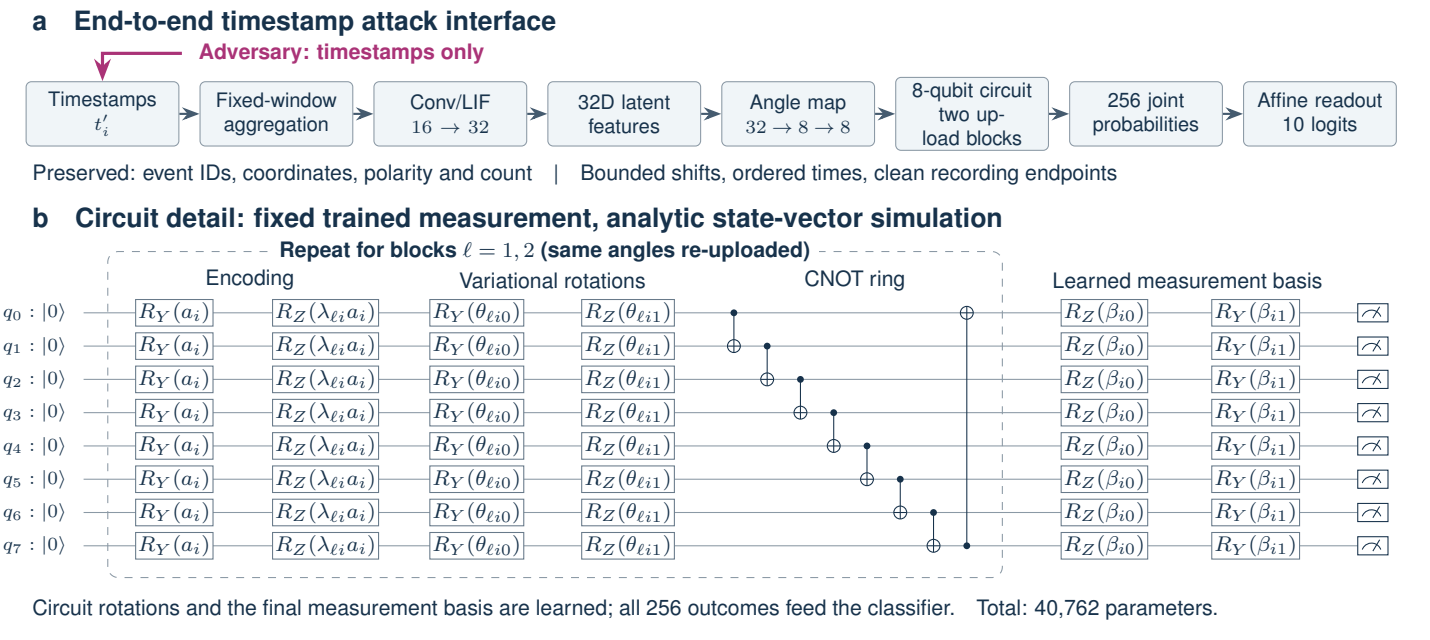

a End-to-end timestamp attack interface
Adversary: timestamps only
Timestamps t'_i
Fixed-window aggregation
Conv/LIF $16 \to 32$
32D latent features
Angle map $32 \to 8 \to 8$
8-qubit circuit two up-load blocks
256 joint probabilities
Affine readout 10 logits
Preserved: event IDs, coordinates, polarity and count | Bounded shifts, ordered times, clean recording endpoints
b Circuit detail: fixed trained measurement, analytic state-vector simulation
Repeat for blocks $\ell = 1, 2$ (same angles re-uploaded)
Encoding
Variational rotations
CNOT ring
Learned measurement basis
$q_0 : |0\rangle$
$q_1 : |0\rangle$
$q_2 : |0\rangle$
$q_3 : |0\rangle$
$q_4 : |0\rangle$
$q_5 : |0\rangle$
$q_6 : |0\rangle$
$q_7 : |0\rangle$
$R_Y(a_i)$
$R_Z(\lambda_{\ell i} a_i)$
$R_Y(\theta_{\ell i 0})$
$R_Z(\theta_{\ell i 1})$
$R_Z(\beta_{i0})$
$R_Y(\beta_{i1})$
Circuit rotations and the final measurement basis are learned; all 256 outcomes feed the classifier. Total: 40,762 parameters.